Completing Two-Column Triangle-Congruence Proofs Answer Key

High School Geometry

Quick Answers

1. —	4. 9	7. —	10. —
2. —	5. —	8. 10	
3. —	6. —	9. —	

Curriculum

Level: High School Geometry
6.EE.B.7 / 7.EE.B.4 – *problems 4, 8*
HSG-GPE.B.4–B.7 – *problems 1, 2, 3, 5, 6, 7, 9, 10*
Difficulty 1–5 of 5 across 10 tagged checks.

Proofs are open reasoning and are marked for human review: accept any logically complete proof, including a different but valid order of statements or an equivalent reason ("Vertical Angles Theorem" for "vertical angles are congruent"). A proof is wrong only if a statement is unsupported, a reason is circular, or the wrong congruence test is named.

1. Given $\overline{AB} \cong \overline{AC}$ and D the midpoint of $\overline{BC}$; prove $\triangle ABD \cong \triangle ACD$. (Two reasons were missing.)
Model proof:

Statements	Reasons
1. $\overline{AB} \cong \overline{AC}$	1. Given
2. D is the midpoint of $\overline{BC}$	2. Given
3. $\overline{BD} \cong \overline{DC}$	3. Definition of midpoint
4. $\overline{AD} \cong \overline{AD}$	4. **Reflexive Property of Congruence**
5. $\triangle ABD \cong \triangle ACD$	5. **SSS**

Why SSS: three pairs of sides were matched and no angle was used. The shared side $\overline{AD}$ is the pair students forget — every proof with a common side needs the reflexive line.

2. E is the midpoint of $\overline{AC}$ and of $\overline{BD}$; prove $\triangle AEB \cong \triangle CED$. (All reasons were missing.)

Model proof:

Statements	Reasons
1. $\overline{AE} \cong \overline{EC}$	1. Definition of midpoint (given: E is the midpoint of $\overline{AC}$)
2. $\angle AEB \cong \angle CED$	2. Vertical angles are congruent
3. $\overline{BE} \cong \overline{ED}$	3. Definition of midpoint (given: E is the midpoint of $\overline{BD}$)
4. $\triangle AEB \cong \triangle CED$	4. SAS

Why SAS and not SSA: the congruent angle at E sits *between* the two pairs of congruent sides, which is what SAS requires.

3. $\overline{PQ} \cong \overline{PR}$ and $\overrightarrow{PS}$ bisects $\angle QPR$; prove $\triangle PQS \cong \triangle PRS$.

Model proof:

Statements	Reasons
1. $\overline{PQ} \cong \overline{PR}$	1. Given
2. $\overrightarrow{PS}$ bisects $\angle QPR$	2. Given
3. $\angle QPS \cong \angle RPS$	3. Definition of angle bisector
4. $\overline{PS} \cong \overline{PS}$	4. Reflexive Property of Congruence
5. $\triangle PQS \cong \triangle PRS$	5. SAS

Note: the included angle is the one the bisector creates, so the order side–angle–side is genuine. Do not assume S is a midpoint — that is a conclusion (by CPCTC), not a given.

4. $\triangle ABC \cong \triangle DEF$ with $AB = 3x - 4$ and $DE = 2x + 5$. Find x.

Solution: $\overline{AB}$ and $\overline{DE}$ are corresponding sides of congruent triangles, so CPCTC gives $AB = DE$:

$$\begin{aligned} 3x - 4 &= 2x + 5 \\ x - 4 &= 5 \end{aligned}$$

$\boxed{x = 9}$ Check: $3(9) - 4 = 23$ and $2(9) + 5 = 23$ ✓

5. $\overline{AB} \parallel \overline{DC}$, E the midpoint of $\overline{AC}$, with B, E, D collinear; prove $\triangle ABE \cong \triangle CDE$.

Model proof:

Statements	Reasons
1. $\overline{AB} \parallel \overline{DC}$	1. Given
2. $\angle BAE \cong \angle DCE$	2. Alternate interior angles (parallel lines cut by transversal $\overline{AC}$)
3. $\overline{AE} \cong \overline{CE}$	3. Definition of midpoint
4. $\angle AEB \cong \angle CED$	4. Vertical angles are congruent
5. $\triangle ABE \cong \triangle CDE$	5. ASA

Why ASA: the congruent side $\overline{AE} \cong \overline{CE}$ lies between the two pairs of congruent angles. (Naming AAS here is not accepted — read the order off the figure.)

6. $\angle W \cong \angle Y$ and $\angle WXZ \cong \angle YZX$; prove $\triangle WXZ \cong \triangle YZX$. (Statements were missing.)
Model proof:

Statements	Reasons
1. $\angle W \cong \angle Y$	1. Given
2. $\angle WXZ \cong \angle YZX$	2. Given
3. $\overline{XZ} \cong \overline{ZX}$	3. Reflexive Property of Congruence
4. $\triangle WXZ \cong \triangle YZX$	4. AAS

Why AAS, not ASA: the shared side $\overline{XZ}$ is *not* between the two marked angles — $\angle W$ is opposite it — so the pattern is angle–angle–side. Note the vertex order in the conclusion: $W \leftrightarrow Y$, $X \leftrightarrow Z$, $Z \leftrightarrow X$.

7. $\overrightarrow{LN}$ bisects $\angle KLM$ and $\angle K \cong \angle M$; prove $\triangle KLN \cong \triangle MLN$.
Model proof:

Statements	Reasons
1. $\overrightarrow{LN}$ bisects $\angle KLM$	1. Given
2. $\angle KLN \cong \angle MLN$	2. Definition of angle bisector
3. $\angle K \cong \angle M$	3. Given
4. $\overline{LN} \cong \overline{LN}$	4. Reflexive Property of Congruence
5. $\triangle KLN \cong \triangle MLN$	5. AAS

Why AAS: $\overline{LN}$ is not between $\angle KLN$ and $\angle K$; it is opposite $\angle K$. Accept ASA only if the student instead pairs $\angle KLN$ with $\angle KNL$, which requires proving those angles congruent first.

8. $\triangle ABC \cong \triangle DEF$ with $m\angle A = 5x + 12$ and $m\angle D = 7x - 8$. Find x.
Solution: $\angle A$ and $\angle D$ correspond, so CPCTC gives $m\angle A = m\angle D$:

$$\begin{aligned} 5x + 12 &= 7x - 8 \\ 20 &= 2x \end{aligned}$$

$\boxed{x = 10}$ Check: $5(10) + 12 = 62$ and $7(10) - 8 = 62$, so $m\angle A = 62°$ in both triangles ✓

9. $\angle A$ and $\angle C$ are right angles and $\overline{AB} \cong \overline{CB}$; prove $\triangle ABD \cong \triangle CBD$, then explain the SSA/HL distinction.
Model proof:

Statements	Reasons
1. $\angle A$ and $\angle C$ are right angles	1. Given
2. $\triangle ABD$ and $\triangle CBD$ are right triangles	2. Definition of right triangle
3. $\overline{AB} \cong \overline{CB}$ (legs)	3. Given
4. $\overline{BD} \cong \overline{BD}$ (hypotenuses)	4. Reflexive Property of Congruence
5. $\triangle ABD \cong \triangle CBD$	5. HL

Model explanation (open — review by hand): SSA fails in general because two different triangles can share the same two sides and non-included angle — the third side can swing to two positions (the ambiguous case). When the known angle is a right angle that ambiguity disappears: the third side is fixed by the Pythagorean theorem, so hypotenuse–leg is enough. HL is SSA with the one angle that makes it safe.

10. Overlapping triangles: $\overline{PS} \cong \overline{PT}$ and $\overline{PQ} \cong \overline{PR}$; prove $\triangle PQT \cong \triangle PRS$ and apply CPCTC.

Model proof:

Statements	**Reasons**
1. $\overline{PQ} \cong \overline{PR}$	1. Given
2. $\angle QPT \cong \angle RPS$	2. Reflexive Property (both name $\angle P$)
3. $\overline{PT} \cong \overline{PS}$	3. Given
4. $\triangle PQT \cong \triangle PRS$	4. SAS
5. $\overline{QT} \cong \overline{RS}$	5. CPCTC

The move that unlocks it: the two triangles overlap and share the angle at P, so that shared angle is the included angle for both. Redrawing $\triangle PQT$ and $\triangle PRS$ separately makes the correspondence $Q \leftrightarrow R$, $T \leftrightarrow S$ visible. CPCTC is the last line, never a reason for congruence itself.